

SHAHKAR AHMAD

AI Engineer | Peshawar

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SUMMARY

Aspiring AI engineer with practical exposure to LLM applications, multi-agent frameworks, and data pipelines. Eager to leverage a foundation in Python, LangGraph, and Scikit-Learn to solve real problems. Seeking an AI/ML Internship opportunity to gain industry experience and refine technical skills.

TECHNICAL SKILLS

AI & LLMs: LangGraph, LangChain, RAG Systems, Multi-Agent Workflows, Human-in-the-Loop (HITL), Prompt Engineering

Machine Learning: Scikit-Learn, TensorFlow, NumPy, Pandas, Matplotlib, Seaborn, EDA, Feature Engineering

Languages & Backend: Python, SQL (PostgreSQL), TypeScript, FastAPI,

Frontend & Tools: Streamlit, React, Tailwind CSS, Git, GitHub, Jupyter Notebook

PROJECTS

Adalat AI Voice-First AI Moot-Court Simulator | [GitHub](#)

2026

- Architected a multi-agent courtroom simulator using a **LangGraph state machine** with a supervisor router orchestrating independent agents (Judge, Opposing Counsel, Witness) backed by two-tier memory.
- Implemented **ReAct-based Judge agents** with dynamic statute lookup tools and Opposing Counsel executing autonomous evidentiary objections verified against the Qanun-e-Shahadat Order 1984.
- Engineered a trustworthy hybrid RAG pipeline fusing **BM25 and dense vector search via RRF ($k = 60$)** with LLM reranking and an automated citation audit engine against an authenticated legal corpus.
- Built a contract-first microservice streaming architecture (Express API, Python FastAPI) supporting real-time STT/TTS, optimizing time-to-first-audio from 16.1s to 6.9s.
- Established a quantitative evaluation harness measuring retrieval performance (MRR 1.00) and multi-agent courtroom behavior across 32 labeled scenarios (Objection F_1 : 1.00, Ground Accuracy: 100%).
- **Tech:** LangGraph, Python, FastAPI, Express.js, React 19, OpenAI API, PostgreSQL, Drizzle ORM

AI Content Factory Automated Multi-Media Generator | [GitHub](#)

2025 – 2026

- Built an automated media generator using a **10-agent LangGraph pipeline** that creates blog posts, social campaigns, podcasts, and videos from a single prompt.
- Implemented parallel processing across worker agents to handle section writing concurrently, reducing overall content generation time.
- Designed a Human-in-the-Loop (HITL) step that pauses execution, allowing users to review and approve content outlines before generation continues.
- Integrated Tavily API for fact-checking/QA evaluation and Gemini 2.5 Flash native audio API to synthesize 3-minute podcast summaries.
- **Tech:** LangGraph, Python, OpenAI API, Gemini API, Tavily, MoviePy, Pexels API

Live Agent Hub AI Streamer Practice Platform | [GitHub](#)

2025

- Built a full-stack platform simulating a live Twitch chat with **8 distinct AI personas** utilizing OpenAI completion endpoints.
- Engineered humanized agent traits including slang integration, typo simulation, chattiness filters, and dynamic screen-context reaction during streams.
- Architected an Express.js backend backed by PostgreSQL/Supabase alongside a modern React/Vite interactive frontend.
- **Tech:** React, Vite, Tailwind CSS, Radix UI, Express.js, TypeScript, PostgreSQL, Supabase, OpenAI API

ChessMastermind Custom Chess AI Engine | [GitHub](#)

2025

- Developed a chess engine from scratch implementing Minimax with Alpha-Beta Pruning, Iterative Deepening, Aspiration Windows, and Quiescence Search.
- Engineered transposition tables (FEN hashing), MVV-LVA move ordering, Killer Move, and History Heuristics for advanced search optimization.
- Created evaluation algorithms analyzing piece-square tables, pawn structure (passed/isolated), king safety, and knight outposts.
- **Tech:** Python, python-chess, FastAPI, Streamlit, NumPy

YouTube Video Summariser | [GitHub](#)

2024

- Built an LLM-powered pipeline extracting YouTube transcripts to generate structured chapter summaries, key insights, and action items.
- **Tech:** LangChain, YouTube Transcript API, GPT-4, Python

EDUCATION

University of Peshawar

BS in Computer Science (AI Specialization)

Oct. 2022 – Expected Oct. 2026

Peshawar, Pakistan

ADDITIONAL ML PROJECTS

Polio Monitors Predictor (2024) ([| GitHub](#)): Forecasted 'Not Available Children' field metrics to optimize campaign resource allocation using Scikit-learn and Pandas.

ODI Run Predictor: Regression model ($R^2 = 0.92$) predicting cricket innings scores from ball-by-ball match data.

Heart Failure Prediction: Classification pipeline evaluating patient clinical metrics for medical risk assessment.

Google Play Store EDA: Exploratory data analysis across 10,000+ Android applications uncovering app dynamics and rating trends.